

2024, Mar. 2024 Release LOC:Acute Adult

Epilepsy

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2, 3)Episode Day 1 ^(1, 3)Episode Day 2 ^(3, 27)Episode Day 3 ^(37, 38)

Episode Day 4

Discharge Screens ⁽⁴⁰⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with known or new onset epilepsy. Seizures related to other underlying issues such as traumatic brain injury, metabolic imbalances, alcohol withdrawal, fever, or other neurologic disorders are excluded and can be found in the Infection: CNS, General Medical, General Trauma, Hypoglycemia, Stroke, and Withdrawal Syndrome subsets. Seizures related to eclampsia are also excluded and can be found in the Hypertensive Disorders of Pregnancy subset.

Introduction:

Epilepsy is a neurologic disorder that is characterized by any of the following:

- At least two unprovoked seizures occurring greater than 24 hours apart
- One unprovoked seizure with at least a 60%(0.60) probability of subsequent seizure(s) occurring within the next 10 years
- Diagnosis of an epilepsy syndrome

Seizures are caused by abnormal excessive or synchronous neuronal activity in the brain and are classified according to level of awareness at seizure onset: focal, generalized, or unknown types; and then classifications are further divided by motor or non-motor symptoms (Scheffer et al., *Epilepsia* 2017, 58: 512-21; Fisher et al., *Epilepsia* 2017, 58: 522-30). Generalized tonic-clonic motor seizures, formerly known as grand mal seizures, can be at seizure onset or secondarily generalized from a focal or nonmotor onset. Current guidelines define the time frame of when a patient is in status epilepticus (SE) differently (Coppler and Elmer, *Critical Care Clinics* 2022, 39: 87-102). The International League Against Epilepsy (ILAE) defines convulsive SE as a seizure lasting greater than 5 minutes, greater than or equal to 10 minutes for focal SE (with or without impaired consciousness), and greater than 10 to 15 minutes for absence seizures (Trinka and Leitinger, *Continuum (Minneapolis)* 2022, 28: 559-602). The Neurocritical Care Society, however, defines SE as greater than or equal to 5 minutes of continuous clinical and/or electrographic seizure activity, or recurrent seizure activity without recovery between seizures (Coppler and Elmer, *Critical Care Clinics* 2022, 39: 87-102). SE is a common neurologic emergency that may require the Critical level of care (Coppler and Elmer, *Critical Care Clinics* 2022, 39: 87-102; Vossler et al., *Epilepsy Curr* 2020, 20: 245-64).

Evaluation and Treatment:

A diagnostic evaluation is usually completed to verify the diagnosis of epilepsy prior to initiating an anticonvulsant medication regimen (Johnson, *Med Clin North Am* 2019, 103: 309-24). Diagnostic testing may include imaging studies, such as magnetic resonance imaging (MRI), computed tomography (CT), or positron emission tomography (PET), traditional or continuous electroencephalogram (EEG), neuropsychological testing, and laboratory studies to rule out

secondary causes.

The goal of epilepsy treatment is to control seizures without causing adverse effects and to improve the quality of life. Seizures can be controlled in 60-70%(0.60-0.70) of patients. Alternative treatments include neurosurgery, ketogenic diet, and vagus nerve stimulation (National Institute for Health and Care Excellence (NICE), *Epilepsies in children, young people and adults*. 2022; Johnson, *Med Clin North Am* 2019, 103: 309-24; Thijs et al., *Lancet* 2019, 393: 689-701).

Note: The frequency of neurological assessment in this subset reflects the opinion of InterQual's® external peer reviewers. Evidence in the medical literature to support the efficacy and effectiveness of the intervention or service is absent, mixed, or unclear. The criteria are based upon current best practice and are the product of an iterative process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁴⁾● Known epilepsy, ≥ **One:**● Post seizure with return to baseline and, ≥ **One:** ⁽⁵⁾– ≥ 2 seizures within 24h and an increase in frequency ^(6, 7)

– Change or progression in seizure type

– Increase in seizure duration

– Postictal state duration > 10 min and < 60 min ⁽⁸⁾● Post seizure, **Both:** ⁽⁹⁾– O₂ sat 90-93%(0.90-0.93) and < baseline ⁽⁵⁾● Intervention, ≥ **One:**

– Oximetry

– Requiring supplemental oxygen ⁽¹⁰⁾● **ACUTE, One:** ⁽¹¹⁾● Epilepsy, **Both:**● Finding, **One:**● Known epilepsy, **One:**– Uncontrolled seizure (excludes status epilepticus) ^(12, 13)– Post seizure with neurological impairment, persistent or unresolved ^(14, 15)– New onset epilepsy ^(6, 16)● Intervention, ≥ **One:**– Continuous EEG ⁽¹⁷⁾– EEG scheduled or performed within 24h ⁽¹⁸⁾– Medication initiation or dose adjustment (includes PO) ^(19, 20)– Neurological assessment at least 3x/24h ⁽²¹⁾– Epilepsy evaluation, scheduled admission ^(22, 23)● Post seizure, **All:** ⁽⁹⁾– O₂ sat ≤ 89%(0.89) and < baseline ⁽⁵⁾

– Oximetry

– Requiring supplemental oxygen ⁽¹⁰⁾● Pregnancy (excludes eclampsia) and, ≥ **One:**

– ≥ 1 seizure within the last 24h

– Postictal state within the last 24h ⁽⁸⁾● **INTERMEDIATE, ≥ One:** ⁽²⁴⁾– HFNC and respiratory monitoring every 3-4h ⁽¹⁰⁾– Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁰⁾● **CRITICAL, ≥ One:** ⁽²⁵⁾● Status epilepticus and, ≥ **One:** ⁽²⁶⁾

– > 2 medications for seizure termination within 24h

– Anticonvulsant, continuous

– HFNC and respiratory monitoring every 1-2h ⁽¹⁰⁾

– Mechanical ventilation

● **Episode Day 1, One:** ^(1, 3)

(Symptom or finding within 24h)

● **OBSERVATION, One:** ⁽⁴⁾● Known epilepsy, **Both:**● Post seizure with return to baseline and, ≥ **One:** ⁽⁵⁾– ≥ 2 seizures within 24h and an increase in frequency ^(6, 7)

– Change or progression in seizure type

- Increase in seizure duration
- **Intervention, ≥ One:**
 - Anticonvulsant (includes PO) initiation or dose adjustment ^(19, 20)
 - CNS imaging scheduled or performed within 24h
 - EEG scheduled or performed within 24h ⁽¹⁸⁾
- Postictal state duration > 10 min and < 60 min ⁽⁸⁾
- **Post seizure, Both:** ⁽⁹⁾
 - O₂ sat 90-93%(0.90-0.93) and < baseline ⁽⁵⁾
- **Intervention, ≥ One:**
 - Oximetry
 - Requiring supplemental oxygen ⁽¹⁰⁾
- **ACUTE, One:** ⁽¹¹⁾
 - **Epilepsy, Both:**
 - **Finding, One:**
 - **Known epilepsy, One:**
 - Uncontrolled seizure (excludes status epilepticus) ^(12, 13)
 - Post seizure with neurological impairment, persistent or unresolved ^(14, 15)
 - New onset epilepsy ^(6, 16)
 - **Intervention, ≥ One:**
 - Continuous EEG ⁽¹⁷⁾
 - EEG scheduled or performed within 24h ⁽¹⁸⁾
 - Medication initiation or dose adjustment (includes PO) ^(19, 20)
 - Neurological assessment at least 3x/24h ⁽²¹⁾
 - Epilepsy evaluation, scheduled admission ^(22, 23)
- **Post seizure, All:** ⁽⁹⁾
 - O₂ sat ≤ 89%(0.89) and < baseline ⁽⁵⁾
 - Oximetry
 - Requiring supplemental oxygen ⁽¹⁰⁾
- **Pregnancy (excludes eclampsia) and, ≥ One:**
 - ≥ 1 seizure within the last 24h
 - Postictal state within the last 24h ⁽⁸⁾
- **INTERMEDIATE, ≥ One:** ⁽²⁴⁾
 - HFNC and respiratory monitoring every 3-4h ⁽¹⁰⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁰⁾
- **CRITICAL, ≥ One:** ⁽²⁵⁾
 - **Status epilepticus and, ≥ One:** ⁽²⁶⁾
 - > 2 medications for seizure termination within 24h
 - Anticonvulsant, continuous
 - HFNC and respiratory monitoring every 1-2h ⁽¹⁰⁾
 - Mechanical ventilation
- **Episode Day 2, One:** ^(3, 27)
 - **OBSERVATION, One:**
 - **Responder, discharge expected today if clinically stable last 12h, All:** ⁽²⁸⁾
 - O₂ sat within acceptable limits
 - No seizure activity or baseline ⁽⁵⁾
 - Postictal state resolved or not present during admission ⁽⁸⁾
 - **Non-responder, not clinically stable for discharge and exceeds initial 24h Observation for this condition, One:**
 - For epilepsy use Episode Day 2 at a higher level of care
 - For a condition other than epilepsy use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Early responder, discharge expected today if clinically stable last 12h, All:** ⁽²⁸⁾

- Afebrile
- EEG monitoring completed or not indicated
- No seizure activity or baseline ⁽⁵⁾
- O₂ sat within acceptable limits
- Postictal state resolved or not present during admission ⁽⁸⁾
- No complication or active comorbidity relevant to this episode of care ⁽²⁹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Continuous EEG ⁽¹⁷⁾
 - Seizure requiring anticonvulsant adjustment (includes PO) ≤ 2d ⁽¹⁹⁾
- **Pneumonia confirmed by imaging and, Both:** ⁽³⁰⁾
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽³¹⁾
 - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽³²⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽⁵⁾
 - WBC ≥ 12,000/cu.mm(12x10⁹/L)
 - WBC ≤ 4,000/cu.mm(4 x10⁹/L)
 - Bands > 10%(0.10)
 - ANC < 500/cu.mm(500x10⁶/L) ⁽³³⁾
 - **Intervention, ≥ One:** ⁽³⁴⁾
 - Requiring supplemental oxygen ⁽¹⁰⁾
 - Anti-infective (includes PO)
 - Respiratory interventions at least 6x/24h ⁽³⁵⁾
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Seizure requiring anticonvulsant and neurological assessment every 3-4h
 - HFNC and respiratory monitoring every 3-4h ⁽¹⁰⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁰⁾
- **CRITICAL, ≥ One:**
 - **Seizure requiring anticonvulsant and neurological assessment every 1-2h and, ≥ One:** ⁽³⁶⁾
 - Continuous
 - Bolus
 - Titration
 - HFNC and respiratory monitoring every 1-2h ⁽¹⁰⁾
 - Mechanical ventilation
- **Episode Day 3, One:** ^(37, 38)
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁸⁾
 - Afebrile
 - EEG monitoring completed or not indicated
 - No seizure activity or baseline ⁽⁵⁾
 - Postictal state resolved or not present during admission ⁽⁸⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²⁹⁾
 - **Complication or comorbidity and clinically stable for discharge, Both:**
 - WBC within acceptable limits ⁽³⁹⁾
 - O₂ sat within acceptable limits ⁽³⁹⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Continuous EEG ⁽¹⁷⁾
 - Seizure requiring anticonvulsant adjustment (includes PO) ≤ 2d ⁽¹⁹⁾
 - **Pneumonia confirmed by imaging and, Both:** ⁽³⁰⁾
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽³¹⁾
 - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 ⁽³²⁾

- O₂ sat ≤ 91%(0.91) and < baseline ⁽⁵⁾
- WBC ≥ 12,000/cu.mm(12x10⁹/L)
- WBC ≤ 4,000/cu.mm(4 x10⁹/L)
- Bands > 10%(0.10)
- ANC < 500/cu.mm(500x10⁶/L) ⁽³³⁾
- **Intervention, ≥ One:** ⁽³⁴⁾
 - Requiring supplemental oxygen ⁽¹⁰⁾
 - Anti-infective (includes PO)
 - Respiratory interventions at least 6x/24h ⁽³⁵⁾
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Seizure requiring anticonvulsant and neurological assessment every 3-4h
 - HFNC and respiratory monitoring every 3-4h ⁽¹⁰⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁰⁾
- **CRITICAL, ≥ One:**
 - Seizure requiring anticonvulsant and neurological assessment every 1-2h and, ≥ **One:** ⁽³⁶⁾
 - Continuous
 - Bolus
 - Titration
 - HFNC and respiratory monitoring every 1-2h ⁽¹⁰⁾
 - Mechanical ventilation
- **Episode Day 4, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽²⁸⁾
 - Afebrile
 - EEG monitoring completed or not indicated
 - No seizure activity or baseline ⁽⁵⁾
 - Postictal state resolved or not present during admission ⁽⁸⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²⁹⁾
 - **Complication or comorbidity and clinically stable for discharge, Both:**
 - WBC within acceptable limits ⁽³⁹⁾
 - O₂ sat within acceptable limits ⁽³⁹⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, One: ⁽⁴⁰⁾

- **HOME, Both:**
 - **Level of care appropriateness, Both:**
 - Home environment safe and accessible ⁽⁴¹⁾
 - Patient or caregiver demonstrates ability to manage care

- **Epilepsy discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁴²⁾
- **Education on the importance of, All:**
 - Strict medication adherence
 - Avoiding activities with the potential for seizure related injury ⁽⁴³⁾
 - Avoiding activities that trigger seizures ⁽⁴⁴⁾
 - Follow up anticonvulsant drug level testing, if applicable
- Medication reconciliation ⁽⁴⁵⁾
- Follow-up care planned ⁽⁴⁶⁾
- Identify and address transportation needs
- Educate caregiver on steps of seizure first aid and when to call 911
- **HOME CARE, Both:**
 - **Level of care appropriateness, All:**
 - Home environment safe and accessible ⁽⁴¹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(47, 48)
 - **Skilled services, ≥ One:**
 - Skilled nursing ⁽⁴⁹⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁵⁰⁾
 - **Epilepsy discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁴²⁾
 - **Education on the importance of, All:**
 - Strict medication adherence
 - Avoiding activities with the potential for seizure related injury ⁽⁴³⁾
 - Avoiding activities that trigger seizures ⁽⁴⁴⁾
 - Follow up anticonvulsant drug level testing, if applicable
 - Medication reconciliation ⁽⁴⁵⁾
 - Follow-up care planned ⁽⁴⁶⁾
 - Identify and address transportation needs
 - Educate caregiver on steps of seizure first aid and when to call 911
- **BEHAVIORAL HEALTH, Both:**
 - **Level of care appropriateness, One:**
 - Support systems available ⁽⁵¹⁾
 - **Skilled treatment, ≥ One:**
 - Clinical assessment ⁽⁵²⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁵³⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁵⁴⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(55, 56)
 - Functional impairments requiring at least supervision ⁽⁵⁷⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴⁵⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

● SUBACUTE MEDICAL OR THERAPY, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ One:

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁵⁴⁾

● Therapy, All:

- Goal directed therapy with rehabilitation potential ^(55, 56)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁷⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁴⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE REHAB, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● Services, All:

- Goal directed therapy with rehabilitation potential ^(55, 56)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁷⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁴⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● ACUTE REHAB, Both:**● Level of care appropriateness, Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, All:

- Goal directed therapy with rehabilitation potential ^(55, 56)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁷⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁵⁸⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁴⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● LONG TERM ACUTE CARE, Both:**● Level of care appropriateness, Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁵⁹⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(60, 61)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁶²⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽⁴⁵⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
100 SEIZURES WITH MCC	n/a	4.5	CMS GMLOS
101 SEIZURES WITHOUT MCC	n/a	2.7	CMS GMLOS
EPILEPSY	32%	3.8	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Guideline Standard Considerations

- Neurological assessment
- Seizure precautions

Expected progress after 24h

- Cessation of seizure activity
- Vital signs stable and within acceptable limits
- Activity increasing

Consider:

- Changing medications to PO
- Discharge planning

4:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

5:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

6:

Instruction: This criteria point may be selected for seizures witnessed in the emergency department (ED), by providers in the hospital setting, or those described pre-hospital by clinicians or non-clinicians that result in the provider ordering the required interventions. The decision to provide these interventions highlights a clinical determination that the events described were seizures and not another condition that mimics a seizure (e.g., hypoglycemia, stroke, transient ischemic attack, syncope).

7:

InterQual® external peer reviewers agree that patients who experience more than two seizures within 24 hours require further intervention in a hospital setting.

8:

Postictal period refers to a variable time interval following a seizure, usually characterized by impaired consciousness (excludes lethargy), and is sometimes also marked by self-limited focal paralysis (Subota et al., Epilepsy Behav 2019, 94: 243-51). Following a seizure, most neurological deficits should resolve within 60 minutes. If deficits persists longer than 60 minutes, nonconvulsive status epilepticus should be suspected (Pottkmper et al., Epilepsia 2020, 61: 1045-61).

9:

Instruction: Selection of this criteria is appropriate for patients with known epilepsy or with a new diagnosis of epilepsy who have an acute oxygen desaturation post seizure.

10:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21))

with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

11:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

12:

Instruction: Application of this criteria point is appropriate for patients with continued seizure(s) despite post initial treatment.

13:

Uncontrolled seizure refers to greater than or equal to 1 seizure(s) post initial treatment.

14:

Seizure related neurological impairment may include:

- Altered mental status (behavior or cognition)
- Ataxic gait
- Postictal paresis/paralysis (e.g., Todd's paresis/paralysis)
- Intractable or incapacitating postictal headache/migraine

15:

Seizure related neurological impairment may include (Galizia and Faulkner, Clin Med (Lond) 2018, 18: 409-13; Wheatley and Ross, Emerg Med Clin North Am 2017, 35: 603-23):

- Altered mental status (behavior or cognition)
- Postictal paresis/paralysis (e.g., Todd's paresis/paralysis)
- Intractable or incapacitating postictal headache/migraine

16:

Epilepsy may be diagnosed after a first time unprovoked seizure if there is a greater than 60%(0.60) probability of subsequent seizure occurring within the next 10 years (Fisher et al., *Epilepsia* 2017, 58: 522-30). A seizure occurring greater than 7 days post-stroke or post-traumatic brain injury (TBI) is considered unprovoked. The timeframe is increased to greater than 1 month for a seizure post-traumatic brain injury with subdural hematoma to be considered unprovoked (Feyissa et al., *Eur J Neurol* 2019, 26: 18-e3; Galizia and Faulkner, *Clin Med (Lond)* 2018, 18: 409-13).

17:

Instruction: This criterion refers to a long-term, continuous electroencephalogram (EEG) which may last less than or equal to 72 hours and may or may not include video.

18:

Instruction: This criterion refers to a 20-30 minute routine (standard) electroencephalogram (EEG) or a prolonged EEG lasting 1-2 hours which may or may not include video.

19:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

20:

Change in medication includes holding a dose if drug toxicity is known or suspected.

21:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

22:

Patients with epilepsy may require a planned, elective admission to the epilepsy monitoring unit (EMU) for continuous electroencephalography (EEG) and video monitoring. EMU length of stay varies based on the reason for admission and seizure frequency. Admission to the EMU may be indicated for (Shih et al., *Epilepsia* 2018, 59: 27-36):

- Classifying seizure type
- Determining seizure frequency
- Differential diagnosis of epileptic seizures versus other etiologies
- Differentiating between seizures versus medication side effects
- Medication adjustment
- Seizure localization as part of a presurgical evaluation

23:

Epilepsy evaluation admission may be indicated for:

- Classifying seizure type
- Determining seizure frequency
- Differential diagnosis of epileptic seizures versus other etiologies
- Differentiating between seizures versus medication side effects

- Medication adjustment
- Seizure localization as part of a presurgical evaluation

24:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

25:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

26:

Current guidelines define the time frame of when a patient is in status epilepticus (SE) in varying ways (Coppler and Elmer, Critical Care Clinics 2022, 39: 87-102). The International League Against Epilepsy (ILAE) defines convulsive SE as a seizure lasting greater than 5 minutes, greater than or equal to 10 minutes for focal SE (with or without impaired consciousness), and greater than 10 to 15 minutes for absence seizure SE (Trinka and Leitinger, Continuum (Minneapolis) 2022, 28: 559-602). The Neurocritical Care Society, however, defines SE as greater than or equal to 5 minutes of continuous clinical and/or electrographic seizure activity, or recurrent seizure activity without recovery between seizures (Coppler and Elmer, Critical Care Clinics 2022, 39: 87-102). SE is a neurological emergency with high mortality and a potential for long-term consequences (Coppler and Elmer, Critical Care Clinics 2022, 39: 87-102; Vossler et al., Epilepsy Curr 2020, 20: 245-64; Strein et al., J Clin Med 2019, 8:).

The focus of treatment is early seizure termination to prevent neurologic sequelae. Guidelines suggest benzodiazepines are the initial therapy of choice in the treatment of patients in SE. Other suggested medications include levetiracetam, phenytoin, fosphenytoin, phenobarbital, valproic acid, midazolam, propofol, thiopental sodium, or pentobarbital (Billington et al., J Clin Med 2016, 5; Glauser et al., Epilepsy Curr 2016, 16: 48-61).

27:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

28:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

29:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

30:

Instruction: This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not

considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

31:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

32:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

33:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection. To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu. mm} (1.5 \times 10^9/L)$

$= 330/\text{cu. mm} (330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

34:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

35:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

36:

Instruction: Medication administration includes continuous infusion, bolus, and titration of medications. For medications that are titrated, the rate and dose are adjusted based on clinical monitoring and laboratory results.

37:**Expected progress after 48-72h**

- Seizure free
- Activities at baseline
- Therapeutic drug levels where feasible
- Stable vital signs preparing for discharge within the next 24 hours

If no, consider:

- Neurology consultation
- Increasing medication dosage for sub-therapeutic levels
- Anticonvulsant regimen modification
- Initiation of Ketogenic diet

38:**Care Facilitation****Potential risk for readmission**

- Knowledge deficit or non-adherence with condition management identified (**Home care, Readmission Reduction Program**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers

- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (i.e., blood lead level testing and follow up)(Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service needs (i.e., serious mental illness, history of hospitalizations, functional impairment), consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

39:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

40:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

41:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

42:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

43:

Caution should be used when performing activities that could result in injury if a seizure occurs. These activities include driving, swimming, bathing, cooking or working with fire, using power tools or equipment, or ascending heights (National Institute for Health and Care Excellence (NICE), Epilepsies in children, young people and adults. 2022).

44:

Patients with epilepsy should avoid activities that trigger seizures such as alcohol or drug use, strobe or flashing lights, high contrast television, sleep deprivation, or electronic screen games (Billington et al., J Clin Med 2016, 5:).

45:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

46:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior

to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

47:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

48:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

49:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

50:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the

need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

51:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

52:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

53:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

54:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

55:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

56:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

57:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

58:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

59:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

60:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

61:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases

- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

62:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.